AREON

WEBSITE & DASHBOARD DESIGN PLAN

Developer m00ny007 — Frontend	**Tech Stack** React (or HTML/CSS/JS), Chart.js, PapaParse
Hosting GitHub Pages (Free)	**Visual Direction** NASA Mission Control / Sci-Fi Terminal

MISSION STATEMENT
Survive Mars. Balance the Power. Read the Data.

A two-page web experience: a single-scroll landing page that pitches the AREON project, plus a separate Mission Control dashboard that demonstrates the data behind the training scenarios.

DESIGN DIRECTION

Dark interface • Mars-red accents • Clean sans-serif typography • Clear data provenance • Responsive layout

Project Overview

The website has two connected destinations:

• **Page 1 — Landing Page:** A single-page scroll experience designed to explain the project quickly and make the gameplay, educational value, and data sources easy for judges to understand.

• **Page 2 — Mission Control Dashboard:** A separate page or tab that visualizes the curated Mars weather and solar-flare training data used by the game.

Visual identity

Use a NASA mission-control / sci-fi terminal aesthetic. Keep the interface dark and readable, with Mars-red highlights and restrained status colors. Prioritize legibility over decorative effects.

Suggested CSS color variables

Token	Hex	Use
Mars Red	#C1440E	Primary accent, alerts, key actions
Dark Gray	#1A1A1A	Main background
Terminal Green	#00FF00	Safe / normal status
Warning Yellow	#FFCC00	Caution / warning status

Page 1 — Landing Page: The Pitch

1. Hero Section — Above the Fold

• **Visual:** Wide, cinematic screenshot from the Unreal game showing the Mars landscape and Base Dome.

• **Title:** AREON

• **Subtitle:** Junior Astronaut Mission Trainer

• **Tagline:** “Survive Mars. Balance the Power. Read the Data.”

Three primary judge-facing actions

■ **WATCH**	90-second gameplay video
■ **PLAY**	Browser trade-off demo
■ **EXPLORE**	Mission Control Dashboard

2. The Problem & Solution — Why AREON?

Space STEM content can feel abstract. AREON gives children ages 7–12 a hands-on way to engage with Mars by managing life-support systems and making mission decisions.

CORE PILLAR
“Every action costs Power, and Power is the only thing keeping you alive.”

3. The Mechanics — How Kids Play

• **HUD:** Show three key resource bars — O■, Power, and Shield.

• **The Trade-off:** Downloading data costs 15 Power. Recharging the shield costs 10 Power. The player must prioritize because both actions cannot be completed with the same limited supply.

• **The Map:** A 1 km Mars terrain containing the Base Dome, Research Center, and Generator, with a five-minute daylight window.

• **The Weather:** Storm conditions are linked to atmospheric pressure drops and solar-flare events in the curated training data.

4. The Three-Day Mission Loop

Scenario	Conditions	Gameplay Effect
Sol 800	Clear skies	Learn the core mechanics
Sol 852	Storm + C-class flare	Shield drains 2×
Sol 2085	Severe global storm + X-class flare	Black skies; Shield drains 2×; O■ drains 1.5×

Landing Page — NASA Data & Transparency

5. NASA Data Integration — Data Validity

Use this section to explain which datasets inform the experience and how they are used. Keep each source's role precise and avoid implying that the game receives live telemetry if it does not.

Source	Use in AREON
Curiosity REMS (PDS)	Historical Mars atmospheric pressure data, including the pressure_pa field.
NASA DONKI	Historical solar-flare event information, including C- and X-class events.
NASA OSDR	Astronaut health context for the AstroGuru AI feature.

Raw CSV evidence

Display the exact four-line CSV excerpt used by the project in a code-style block. Replace the placeholder below with the verified contents of the project's actual weather_data.csv file; do not type or reconstruct values from memory.

```
weather_data.csv
[Insert the exact header row]
[Insert verified data row 1]
[Insert verified data row 2]
[Insert verified data row 3]
```

6. Data Honesty — Compliance & Transparency

Recommended public-facing disclosure:

> "Because a perfect three-day consecutive storm does not exist in the archive, we curated three specific historical Sols (800, 852, 2085) to act as escalating training scenarios. Solar flares were mapped to these days for gameplay. This is a curated training dataset, not live telemetry."

7. Team & Compliance

• **Unmesh:** Lead

• **Sudharsan:** Systems

• **Dilli:** Level Design

• **m00ny007:** Frontend

• **Compliance:** Combat-free; anonymous submission using AI voice; built during the 48-hour November 14–15 sprint.

Page 2 — Mission Control Dashboard

Purpose

The dashboard should look like a focused telemetry screen. It demonstrates that the game's weather scenarios are data-driven and lets visitors inspect and download the same CSV file used by the game.

1. Live Graph — Chart.js

• **X-axis:** Sol values 800, 852, and 2085.

• **Y-axis:** Atmospheric pressure in pascals (Pa).

• **Data points:** 887 Pa, 924 Pa, and 772 Pa, respectively, as supplied in this plan.

• **Threshold:** A bold horizontal reference line at 800 Pa.

• **Animation:** Draw the line from left to right. Use a restrained warning highlight when the selected scenario is below the pressure threshold or has an associated flare.

2. Live Metrics

Metric	Display	Suggested status treatment
Current Pressure	Selected Sol's pressure value in Pa	Numeric readout
Current Flare Status	NONE / C / X	Green / Yellow / Red
Storm Status	SAFE / WARNING	Neutral / warning

3. Data Source Footer & CSV Download

• Footer text: "Data provided by NASA Curiosity REMS (PDS) and DONKI. Curated for AREON. Not live telemetry."

• Provide a clearly labeled button that downloads the actual weather_data.csv file directly from the website.

Dashboard behavior logic

The following is the intended warning rule. The flare value should be normalized consistently in the CSV (for example: none, C, X). Confirm whether the pressure threshold is a gameplay rule or only a visual reference before treating it as a scientific storm classification.

```
if (pressure < 800 || flare !== 'none') {
  document.body.classList.add('storm-warning');
}
```

Important implementation note

A page-wide red flash can be distracting and may reduce accessibility. Prefer a brief, subtle warning transition and a persistent status label. Respect reduced-motion preferences and ensure warning states remain understandable without color alone.

Frontend Developer Action Plan

Step 1 — Set up the skeleton

• Create index.html for the landing page and dashboard.html for Mission Control.

• Use CSS variables for the Mars Red, Dark Gray, Terminal Green, and Warning Yellow palette.

• Keep shared navigation and visual styles consistent across both pages.

Step 2 — Build the landing page sections

• Implement the hero with the Unreal gameplay screenshot and the three WATCH / PLAY / EXPLORE actions.

• Embed the 90-second YouTube gameplay video.

• Add the problem/solution, mechanics, three-day mission, NASA data, transparency, and team sections.

• Insert Dilli's map screenshots in the mission section.

Step 3 — Build the dashboard

• Use PapaParse to load and parse the project CSV.

• Use Chart.js to plot pressure against Sol and display the 800 Pa reference line.

• Update pressure, flare, and storm metrics from the selected data row.

• Apply warning styling when the configured condition is met.

• Add a download link to the exact weather_data.csv asset used by the game.

Additional visual request

• Add a slowly rotating Mars globe in the landing page hero section background. Keep the text and primary actions readable above it; use a subtle glow and avoid competing with the gameplay screenshot.

Suggested build order

Priority	Deliverable	Acceptance check
1	Page skeleton + navigation	Both pages load and link to each other
2	Landing page content	All pitch sections are present and readable
3	CSV parsing + dashboard graph	Chart values match the verified CSV
4	Metrics + warning states	Status reflects the selected scenario
5	CSV download + final polish	Downloaded file matches the game's source CSV

Pre-publication checks

• Verify every NASA dataset citation, field name, Sol mapping, and flare classification against the actual source records.

• Make clear that the three Sols are curated historical training scenarios, not consecutive live observations.

• Ensure the displayed CSV excerpt and download are the exact file used by the game.

• Test the site on desktop and mobile; check keyboard access, contrast, and reduced-motion behavior.

• Confirm that all links, video embeds, screenshots, and GitHub Pages paths work after deployment.

End of design plan